Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Christopher Paul Anderson	Email	cpand@illinois.edu
Institution / Department	*University of Illinois Urbana-Champaign* *Materials Science and Engineering, Physics, Electrical and Computer Engineering*			
Address	104 South Goodwin Ave Urbana, IL 61801			

2. Technology Overview

Title of Proposed Technology	*Silicon carbide quantum technology*			
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (√)**, Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), **Quantum (√)**			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		*Electrical and Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	*Semiconducting Materials*	*Optical Application Devices*	*Optical Devices*
Concept of Proposed Technology	*Silicon carbide, a wafer-scale wide bandgap semiconductor, is also one of the leading materials for quantum technologies (1). Defect qubits hosted in this crystal display the longest coherences reported for any qubit candidate, and naturally interface to near infrared light for readout and state preparation (2,3). These systems can find use in nanoscale sensing, competing with the NV center in diamond, but also in distributing quantum entanglement with light. Such optically-mediated entanglement could form the backbone of quantum communications networks as quantum repeaters, or interconnect modular quantum computing systems. In these cases, the wafer scalability of both electronic (4) and photonic circuitry provide a unique advantage.* *PI Anderson is the world-leader in this platform (1-5). We create defect qubits and quantum photonic circuitry, integrate them with traditional CMOS structures, and measure both electron spins and single photons in cryogenic environments. We aim to create both on and off-chip entanglement, mitigate quantum noise sources, discover new qubit candidates, and make better quantum sensors.* *The eventual goal is to enable a scalable internet of quantum devices using SiC quantum photonic systems. This blends traditional semiconductor materials, photonics, and quantum technology.*			

(max 10 pages)

Rationale for Joint R&D Project	*Our research group can measure silicon carbide quantum devices in multiple cryogenic photonic probe stations. We have the necessary single photon detectors, waveform generators, tunable lasers, etc. We make silicon carbide on insulator (SiCOI), an emerging thin film platform for photonics and electronics in house. This involves bonding, CMP, precision lapping. We also have the ability to use atomic layer etching to make ultra-smooth SiC surfaces.* *However, we do not have the ability to grow or produce SiC. This includes the quantum-grade epitaxy needed for quantum applications. Our fabrication is also small scale, and we lack the traditional SiC electronic device technology (i.e. p- and n- doping) needed for scalable quantum devices. The ability to localize defect qubits with ion implantation, passivate surfaces (drawing from the SiC electronics industry) is underdeveloped. In addition, use cases of these technologies for quantum networking and sensing remain unexplored compared to diamond counterparts. Therefore, there is advantage in partnering with Korean SMEs who are experts in one of the following areas and beyond:* • *SiC growth* • *SiC processing (etching, CMOS, heteroepitaxy of nitrides, etc)* • *SiC electronics* *In addition, there is opportunity to harness SiCOI for related technologies in MEMs (SiC is extremely low loss and stiff), photonics (wide bandgap and nonlinear), and electronics. We hope to partner with SMEs interested in those areas.* *We therefore desire a collaboration with semiconductor, photonics, or quantum companies who can help scale this technology, solve outstanding challenges, and explore use cases. We also hope to connect with experts in SiC materials.*
Keywords	1. Single photon sources 2. Silicon carbide 3. Quantum repeaters 4. Photonic devices 5. Spin qubits

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

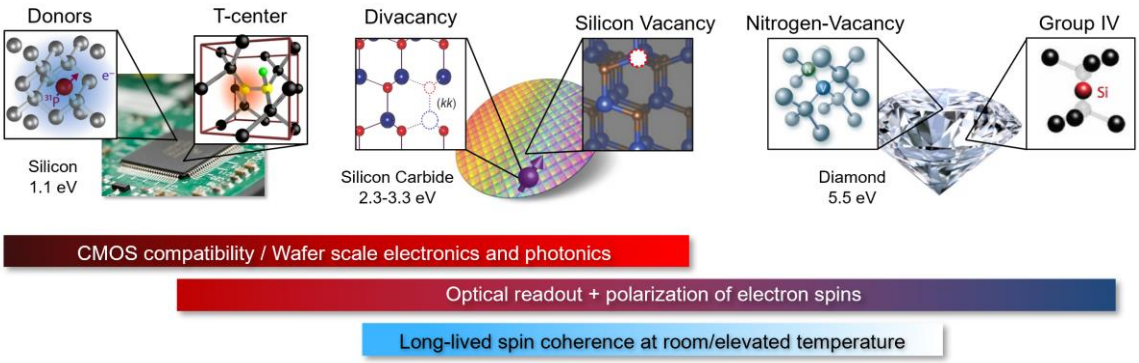

Figure 1: Comparison of solid-state optically active qubit platforms. SiC blends the scalability of silicon with diamond-like properties and qubits.

Our general R&D objectives focus on:

- ***Creation of quantum repeaters.*** *Silicon carbide qubits interfaced with light are natural candidates to create quantum repeaters that scale quantum communications. While almost all requirements have been established in SiC, verification of spin-photon entanglement and remote heralded entanglement are the current focus. Entanglement rates will be increased with nanophotonic devices and advanced packaging.*

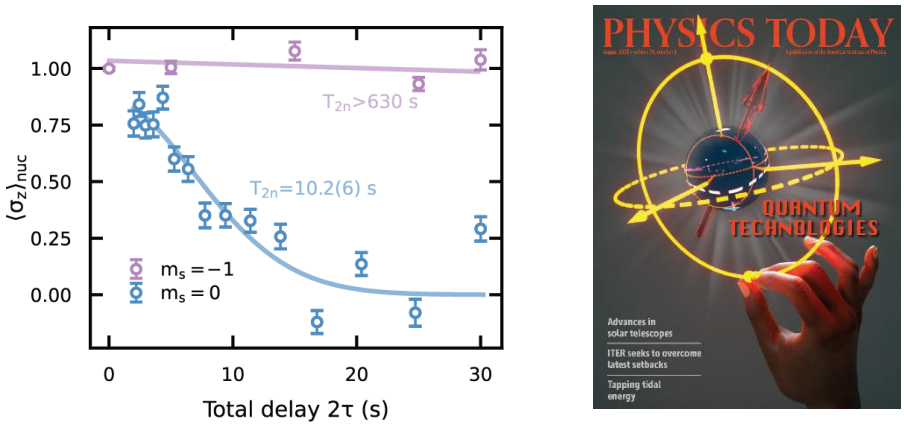

Figure 2: (left) The longest Hahn echo time in the world (3). Single nuclear spins in SiC display T_2 times longer than any atomic, ionic, or superconducting quantum system—over ten minutes. These form robust quantum memories for quantum technologies. (right) The promise of SiC quantum technologies was featured by out cover article in *Physics Today (1)*.

- ***Mitigating noise for qubits.*** *The largest problem for this platform in sensing and communications are the sources of noise in the material (other defects), and uncontrolled surfaces. Our goal is to leverage SiC MOSFET passivation technologies, atomic layer deposition, and materials purification to increase quantum coherence.*
- ***Localization of qubits, new qubit discovery.*** *To realize scalable systems, defects need to be created in nanoscale mode volumes of optical cavities on demand. This requires nanoimplantation or others approaches. In addition, we are developing emerging defect qubit candidates in SiC which emit in the telecom C-band for optimal fiber transmission.*

(max 10 pages)

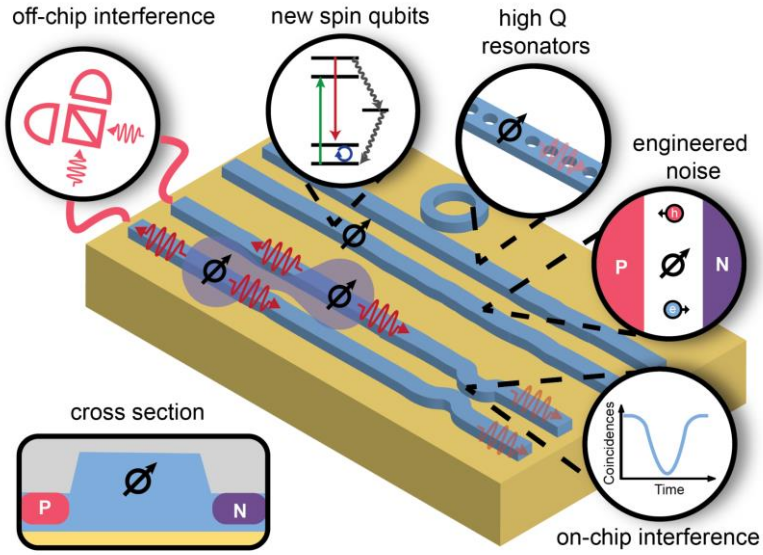

Figure 3: Current and potential future research directions. SiCOI photonic integrated circuits with doping control to mitigate noise can create efficient on and off-chip quantum entanglement generation.

- ***Silicon carbide photonic integrated circuits.*** *SiCOI films are created in our lab and are not commercially available. Photonic devices from these materials not only enhance spin-photon entanglement rates, but can also boost sensitivity in quantum sensing protocols. SiCOI photonics also finds use as nonlinear optical elements. Our efforts relate to reducing the TTV and increasing the quality of these films, and exploring electronic + photonic devices similar to SOI photonics. We also aim to integrate functionality like SNSPDs on chip.*
- ***Hybrid entanglement & scaling quantum computers.*** *Through quantum frequency conversion, SiC qubits can interface to quantum computing platforms, facilitating memory-enhanced scaling. This is related to the approach by IonQ. We are currently attempting to entangle a Sr ion trap quantum computing module with SiC defects to create to first solid-state ion entanglement.*

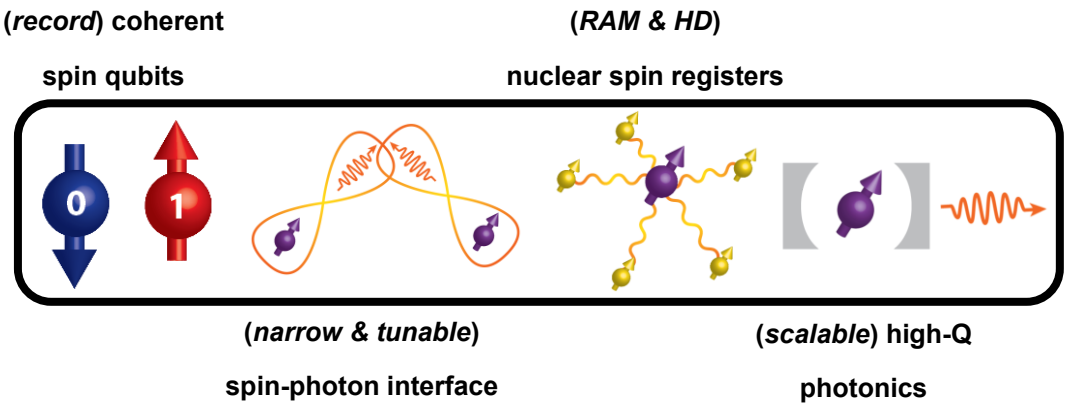

Figure 4: The SiC quantum platform combines record coherent spin qubits, entangled with coherent and tunable single photons, access to robust nuclear spin memory and wafer-scale high-Q photonics.

Overall, we aim to scale and optimize the SiC platform through materials, photonics, noise mitigation for use in quantum technologies. Integration in SiC allows for unique R&D opportunities and openings for large advances in maturity.

(max 10 pages)

R&D Target Specifications	*Through a potential partnership, we hope to:* • *Increase the scalability of SiCOI films with advanced wafer processing, growth, TTV reduction to 100 nm at 100 mm wafer scales.* • *Create high quality photonic structures with Q>1,000,000, <0.3 dB/cm in SiC.* • *Develop CMOS doping capability in SiC compatible with SiCOI and quantum defects.* • *Localize defects with <50 nm precision.* • *Grow heteroepitaxial films of AlN or AlScN on SiCOI* • *Translate SiC electronics processing and treatment to reduce spectral diffusion in films to <100 MHz.* • *Explore use cases in MEMs and electronics for SiCOI* • *Explore use cases of SiC quantum sensing that competes with the NV center.* *These advances will enable scalable SiC quantum technology beyond what is possible in academia.*
R&D Plan	*This is difficult to provide without knowledge of the SME partner.* *For Phase 1, we envision that SiC materials transfer and process development and exchange will occur. These will be validated in PI Anderson's lab in terms of impact on quantum performance.* *In Phase 2, we envision that a feedback loop is created between quantum device fabrication and measurement (UIUC) and materials development/scaling/applications/processing (Korean SME). By the end of the project, wafer scale SiC quantum photonic circuitry will be enabled and a pathway will be adopted for technology transfer to semiconductor, quantum, and photonics industries. A useful quantum prototype will be created that exceeds the state of the art. Patents will be filed and demonstration collaborative publications submitted. We expect adoption and interest from quantum companies.*

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*UIUC will fabricate, measure, and characterize SiC quantum devices and materials. Qubit measurements at cryogenic temperatures will be performed. This will include use of UIUC's nanofabrication facilities and PI Anderson's custom cryogenic and electronic photonic probe stations. Work will be performed by graduate researchers and postdoctoral scholars. UIUC will provide guidance to Korean partners on the broad use of SiC for quantum technologies, which may be of interest to the semiconductor industry. UIUC will publish collaborative research articles.*
Expected R&R of Korean SME Partner	*The following is flexible, given the interests and capabilities of the SME partner.* *We expect the SME potentially provides materials growth, processing, transfer and initial characterization of materials. Materials will be supplied to UIUC as part of the R&D project, not as a paid product. Doping and surface treatments may be performed.* *We expect the research scope to be primarily in optimizing processes to result in optimal quantum performance.* *The SME may also not focus on synthesis, and instead guide UIUC researchers on applications and prototypes using existing efforts in SiC. Other topics related to the R&D Target Specifications are of interest.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Korean SMEs will develop materials and products outside of their normal scope in collaboration with UIUC to capture quantum photonics and quantum technology markets. New science will be enabled, leading to high impact publications. Prototypes and patents will be developed. Commercialization will be explored with UIUC's quantum partners and Korean companies. Quantum networking and sensing sectors will be affected. Korean companies will become key to the emerging high-tech materials for quantum, while research at UIUC will be accelerated by new capabilities. Specifically, large scale processing of SiC quantum devices may be established.*

6. References

(1) "Embracing imperfection for quantum technologies." C. P. Anderson and D. D. Awschalom. Physics Today 76 (8), 26–33 (2023)

(2) "Five-second coherence of a single spin with single-shot readout in silicon carbide." C. P. Anderson*, E. O. Glen*, C. Zeledon, A. Bourassa, Y. Jin, Y. Zhu, C. Vorwerk, A. L. Crook, H. Abe, J. Ul-Hassan, T. Ohshima, N. T. Son, G. Galli, and D. D. Awschalom. Science Advances 8, 5 (2022)

(3) "Minute-long quantum coherence enabled by electrical depletion of magnetic noise" C. Zeledon, B. Pingault, J. C. Marcks, M. Onizhuk, Y. Tsaturyan, Y. Wang, B. S. Soloway, H. Abe, M. Ghezellou, J. Ul-Hassan, T. Ohshima, N. T. Son, F. J. Heremans, G. Galli, C. P. Anderson, D. D. Awschalom. arXiv 2504.13164 (2025) (In review at Science)

(4) "Electrical and optical control of single spins integrated in scalable semiconductor devices." C. P. Anderson*, A. Bourassa*, K. C. Miao, G. Wolfowicz, P. J. Mintun, A. L. Crook, H. Abe, J. Ul-Hassan, N. T. Son, T. Ohshima, D. D. Awschalom, Science 336, 6470, 1225-1230 (2019)

(5) "Entanglement and control of single nuclear spins in isotopically engineered silicon carbide." A. Bourassa*, C. P. Anderson*, K. C. Miao, M. Onizhuk, H. Ma, A. Crook, H. Abe, J. Ul-Hassan, T. Ohshima, N. T. Son, G. Galli, D. D. Awschalom. Nature Materials 19, 1319-1325 (2020)

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

--- 10-page limit ---